NYU

Thomas Sargent <ts43@nyu.edu>

Potential Typo in Quant Econ 64 Eliminating Cross Products

2 messages

Jie Wei <jw7956@nyu.edu> Thu, Oct 26, 2023 at 8:45 PM
To: Thomas Sargent <thomas.sargent@nyu.edu>

Hi Tom,

I hope this email finds you well!

This is not about tomorrow's exam. I think there are two typos in quant econ 64: Eliminating Cross Products.

The problem is to choose $\{x_{t+1}, u_t\}_{t=0}^{\infty}$ to maximize

$$-\sum_{t=0}^{\infty}(x_t' R x_t + u_t' Q u_t + 2 u_t H x_t)$$

subject to the linear constraints

$$x_{t+1} = A x_t + B u_t, \quad t \geq 0$$

where x_0 is a given initial condition.

The solution to this undiscounted infinite-horizon problem is a time-invariant feedback rule

$$u_t = -F x_t$$

where

$$F = -(Q + B'PB)^{-1} B'PA$$

First, the objective function should be ... + 2u'Hx instead of ... + 2uHx, i.e, we should have the transpose of u.

Second, the F here should not be the same as the F without cross product. Here is my attempt to solve for the F, and I think it takes the form (Q+B'PB)^{-1}(H+B'PA), i.e, there is no "-" sign and there is an extra H

Guess value function take the form $x'Px$, then

$$-x'Px = -\max_u [x'Rx + u'Qu + 2u'Hx + (Ax + Bu)'P(Ax + Bu)]$$

Take the first order condition, we get

$$\begin{aligned} 2Qu + 2Hx + 2B'P(Ax + Bu) &= 0 \\ Qu + Hx + B'PAx + B'PBu &= 0 \\ (Q + B'PB)u &= -(H + B'PA)x \\ u &= -(Q + B'PB)^{-1}(H + B'PA)x \end{aligned}$$

which means $F = (Q + B'PB)^{-1}(H + B'PA)$

Best regards,
Jie

Thomas J Sargent <thomas.sargent@nyu.edu> Thu, Oct 26, 2023 at 8:50 PM
Reply-To: thomas.sargent@nyu.edu
To: Jie Wei <jw7956@nyu.edu>

Dear Jie,

Hi. Thanks so much. I will check this tomorrow and then fix it in the quantecon lecture this weekend I hope!

Good luck on the exam.

Best wishes, Tom
[Quoted text hidden]